

Oral Exam (Probability Theory)

Name: Faran Taimoor Butt

Email: faranbutt@phystech.edu

1. (1 point) Definition of conditional Probability.

Solution:

Conditional Probability is the probability of event A given another event B has already occurred.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

where

$P(A \cap B)$: probability of A and B both occurring

$P(B)$: Probability of B occurring

2. (1 point) Definition of Posterior and Prior Probability

Solution: Prior Probability:

$P(A)$ is the prior probability of A.

The initial probability of Probability A. It represents prior belief about A

Posterior Probability:

$P(A|B)$ is the posterior probability of A.

The updated probability of the event A after observing the event B.

3. (1 point) Bayes rule?

Solution: Bayes rule is a way to relate posterior probability, prior probability and likelihood.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

4. (1 point) Law of total probability?

Solution: The law of total probability is an easy way in probability theory that helps us to calculate total probability by breaking it down into smaller parts.

If you have an event B and a set of scenarios from $A_1, \dots, A_n$ that are mutually exclusive and cover all possibilities then

1. A_i is disjoint

2. $A_1 \cup \dots \cup A_n = S$

$$P(B) = \sum_{i=1}^n P(B|A_i)P(A_i)$$

5. (1 point) relation between Law of Total Probability and Bayes rule

Solution: As we know in Bayes rule

$$P(A_k|B) = \frac{P(B|A_k)P(A_k)}{P(B)}$$

We can expand the denominator using Law of Total Probability

$$P(A_k|B) = \frac{P(B|A_k)P(A_k)}{\sum_{i=1}^n P(B|A_i)P(A_i)}$$

where

$P(A_k)$ = prior probability

$P(B|A_k)$ = likelihood

$P(A_k|B)$ = posterior probability

$P(B)$ = marginal likelihood

6. (1 point) Monty Hall solution?

Solution: let

A = Prize behind given door

B = Choice of door was changed

$\tilde{B}$ = Choice of door was not changed

W = Win

$$P(W|B) = P(W|B, A) \cdot P(A) + P(W|B, \tilde{A})P(\tilde{A})$$

$$P(W|B) = 0 \cdot \frac{1}{3} + 1 \cdot \frac{2}{3} = \frac{2}{3}$$

$$P(W|\tilde{B}) = P(W|\tilde{B}, A) \cdot P(A) + P(W|\tilde{B}, \tilde{A})P(\tilde{A})$$

$$P(W|\tilde{B}) = 1 \cdot \frac{1}{3} + 0 \cdot \frac{2}{3} = \frac{1}{3}$$

As

$$P(W|B) = \frac{2}{3} > P(W|\tilde{B}) = \frac{1}{3}$$

user must change the door